

#Inheritance

I. Single Level Inheritance

1. Online Shopping User

Create:

User class → name, email

Customer class → inherits from User and adds cart_items

Task

Write a method in Customer to display full user details including cart count.

2. Banking System (Interview Favorite)

Create:

Account → account_number, balance, method deposit()

SavingsAccount → adds interest_rate and method apply_interest()

Requirements

Use constructor chaining with super(). Show balance before and after interest

II. Multilevel Inheritance

1. Employee Hierarchy

Create a chain:

Employee → Developer → BackendDeveloper

Task

Employee → name

Developer → programming_language

BackendDeveloper → framework

Display all details from child class.

2. Food Delivery App Roles

Create:

Person → DeliveryAgent → SeniorDeliveryAgent

Requirements

Person → name, phone

DeliveryAgent → vehicle_type

SeniorDeliveryAgent → region, bonus calculation method

Task

Implement a method in the lowest class that prints a performance summary.

III. Multiple Inheritance

1. Smart Device

Create:

MusicPlayer → play_music()

Camera → take_photo()

Create SmartDevice inheriting from both and demonstrate both features.

2.E-commerce Seller Dashboard

Create:

Seller → upload_product()

Analytics → view_sales_report()

Create:

class SellerDashboard(Seller, Analytics)

Task

Implement a method that simulates a seller session:

Upload product

Show report

This tests understanding of multiple parent classes and shared object state.

Python allows a class to inherit from multiple parents and combines their features into one derived class.

IV. Hierarchical Inheritance

1.School System

Create:

```
    Person
   /    \
 Student Teacher
Task
Person → name
Student → marks
Teacher → subject
```

Print details for both subclasses.

2.Payment Gateway System

Create:

```
    Payment
   /      \
CardPayment UPIPayment
Requirements
Payment → amount, method process_payment()
```

Subclasses override method with different processing messages
Interview concept tested

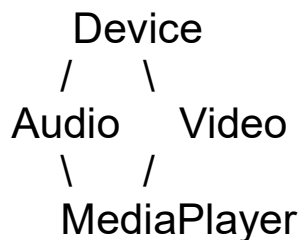
Method overriding in hierarchical structure and polymorphic calls.

Hierarchical inheritance occurs when multiple subclasses inherit from a single base class,
allowing shared functionality across different roles.

V. Hybrid Inheritance (Most Asked in Interviews)

1. Media Player System

Create:



Task

Device → power_on()

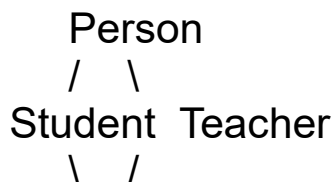
Audio → play_audio()

Video → play_video()

Create a MediaPlayer object and call all methods.

2. University Portal (Diamond Structure)

Create:



TeachingAssistant

Requirements

Use super() in constructors

Ensure attributes from Person are not initialized twice